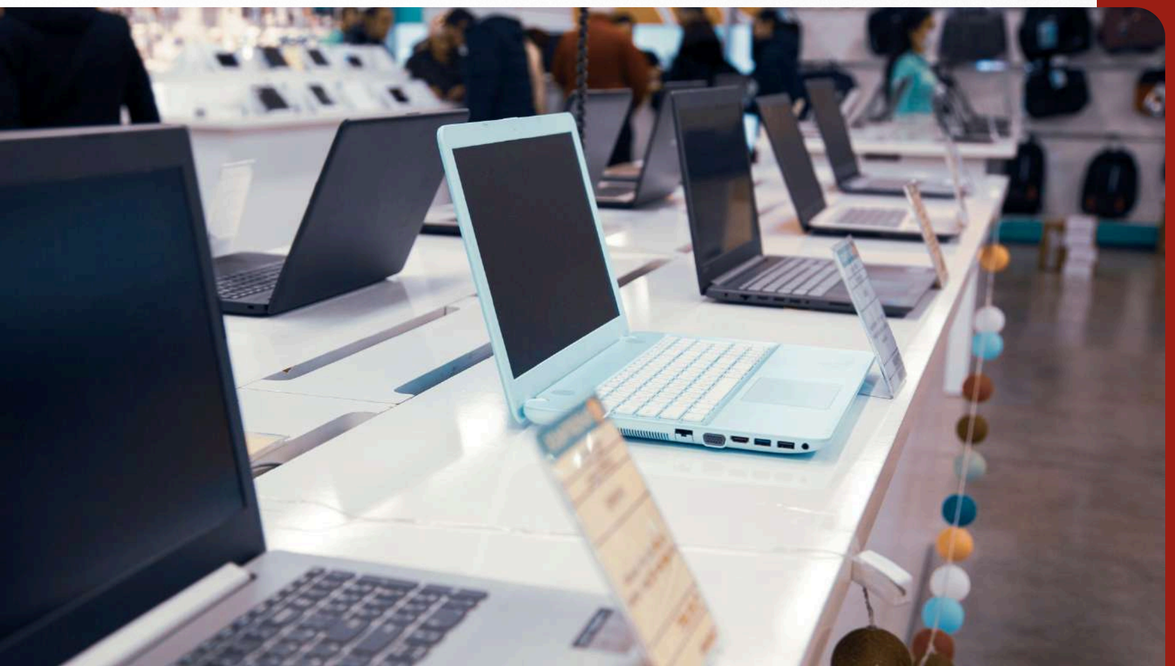
Introduction

This analysis evaluates **2025 Retail Electronics sales performance** using Total Revenue as the key metric, focusing on trends, product tiers, channels, and customer behavior to identify key growth drivers.



01 Which factors drive revenue performance: product tier, sales channel, or transaction volume?

02 Do top products significantly contribute more to revenue than low-value products?

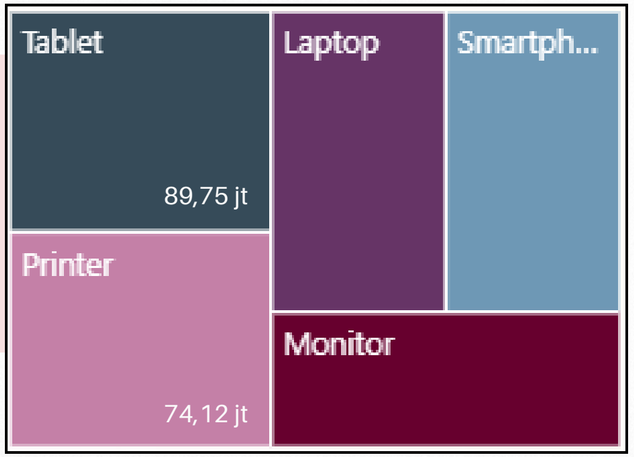
03 Are there meaningful differences across customer segments and provinces?

04 Which time periods show seasonal decline or growth that require strategic action?

Product Tier Impact on Revenue (2025)

Product Dominance

In 2025, **tablets and printers** consistently lead revenue across both channels. These top-tier products generate 21% higher revenue compared to low-tier products, with the difference statistically significant*



* tested with Mann-Whitney U

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                            OLS Regression Results
==============================================================================
Dep. Variable:     np.log(Total_Penjualan)   R-squared:                       0.555
Model:                                 OLS   Adj. R-squared:                  0.552
Method:                      Least Squares   F-statistic:                     206.7
Date:                     Sat, 21 Feb 2026   Prob (F-statistic):           4.72e-87
Time:                             10:00:50   Log-Likelihood:                -474.16
No. Observations:                      502   AIC:                             956.3
Df Residuals:                          498   BIC:                             973.2
Df Model:                                3
Covariance Type:                 nonrobust
==========================================================================================
                             coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------------------
Intercept                 16.6856      0.074    225.575      0.000      16.540      16.831
C(Product_Tier)[T.Low]    -0.1305      0.056     -2.316      0.021      -0.241      -0.020
C(Channel)[T.Online]      -0.0322      0.056     -0.577      0.564      -0.142       0.078
Qty                        0.1251      0.005     24.523      0.000       0.115       0.135
==============================================================================
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Revenue Drivers

- The model explains 55.5% of revenue variation and confirms that **revenue growth is predominantly volume-driven**
- Each additional unit sold **increases revenue by approximately 13%**, while low-tier products generate around 12% less revenue than high-tier products
- Sales channel differences show no significant impact

Customer Segment

High Performing Market

MoM Total 3.84%

Sumatera Utara

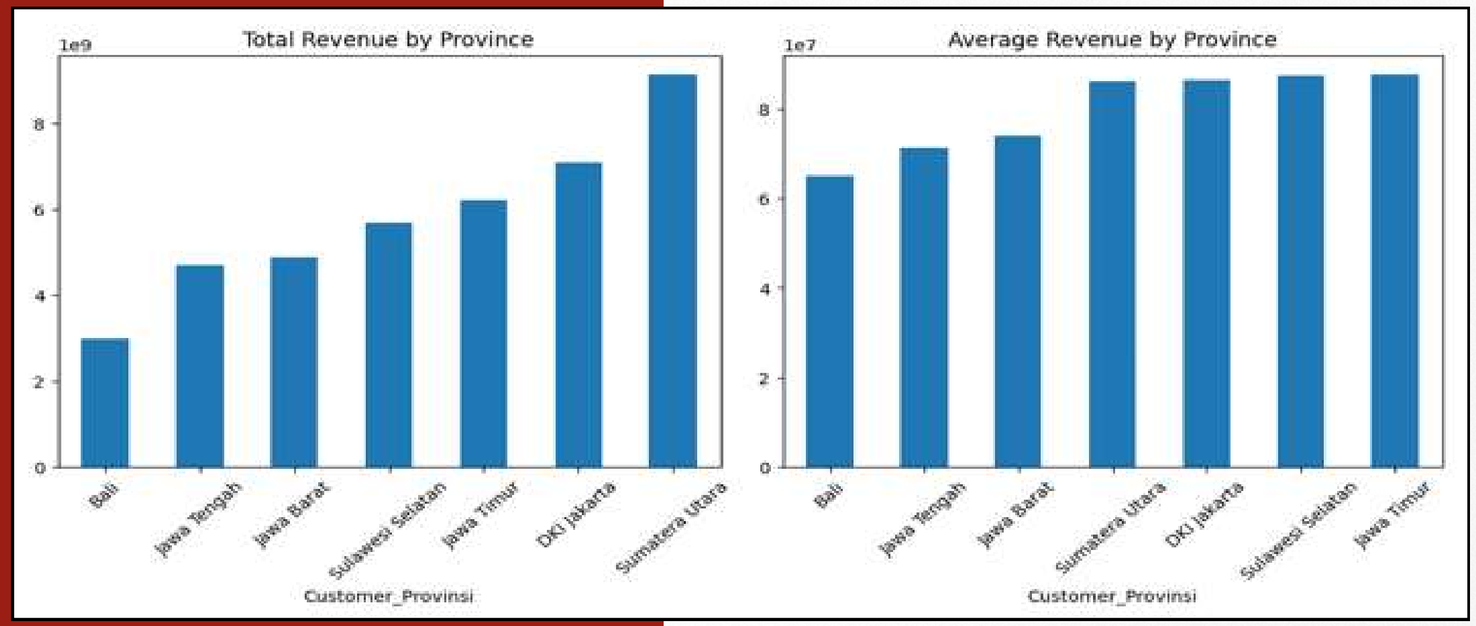
Top Growth Market

MoM Total 11.52%

DKI Jakarta

Statistical analysis shows **no significant differences*** in purchase volume, total revenue, or product tier preference across customer segments, suggesting **no need for segment-specific pricing or targeting strategy**

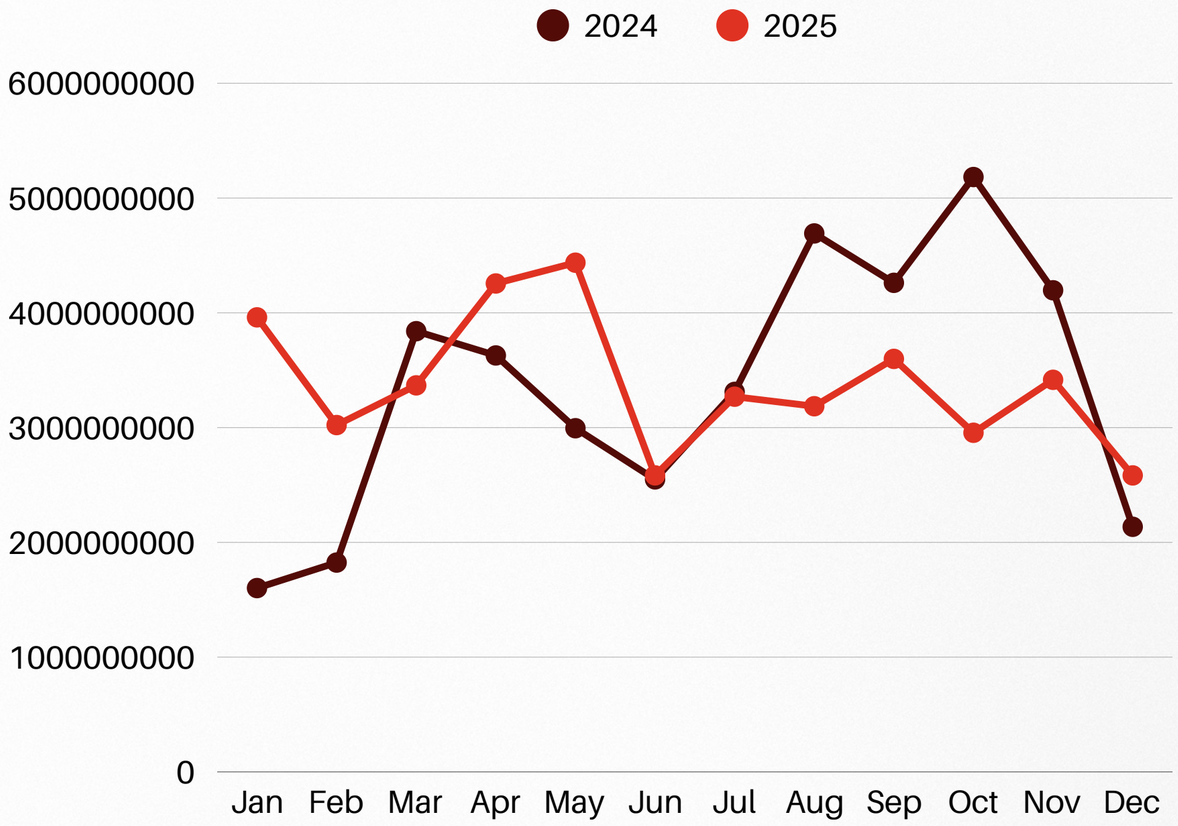
* tested with Kruskal Wallis and Chi-Square

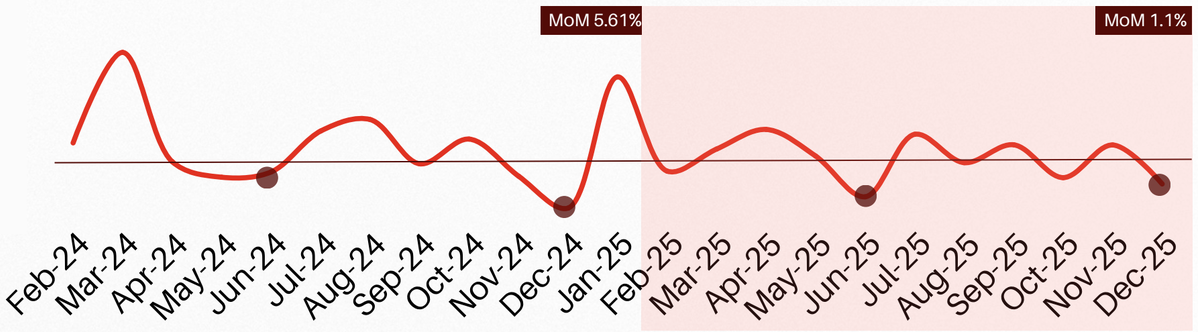


Customer Province

Although North Sumatra leads in both average and total revenue, followed by DKI Jakarta and East Java, **statistical testing shows no province significantly outperforms others**, indicating **broadly comparable market potential across regions**, with Bali operating at a smaller scale.

Revenue Sales Trend





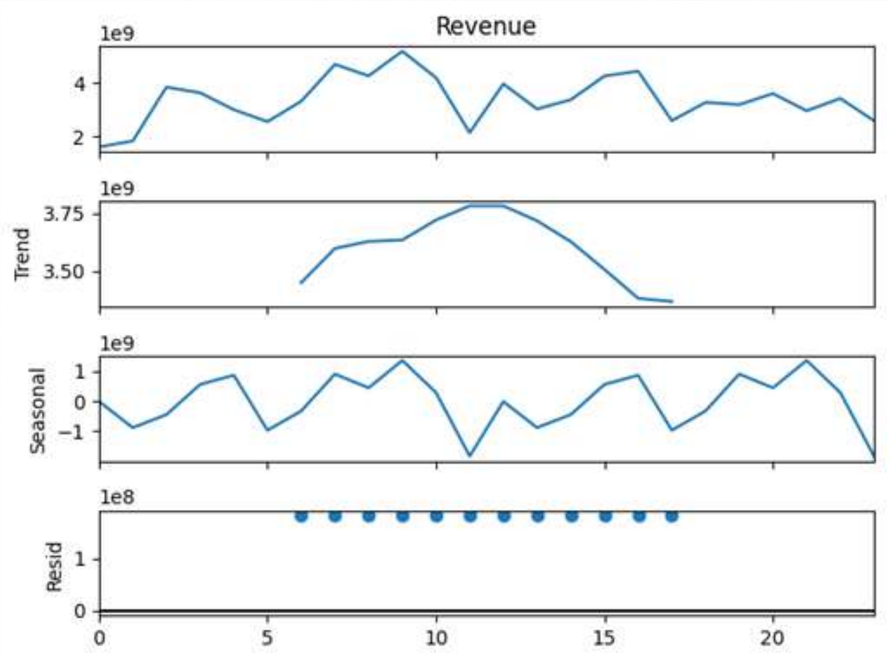
Overview

While 2024 shows stronger revenue peaks in several months, 2025 demonstrates a more stable performance, with a modest overall increase of 1.1% compared to 2024

Low Season Period

Both years reveal a recurring seasonal dip mid-year (Jun-Jul), with performance tapering off again toward year-end (Nov-Des)

Trend Decomposition



High season:

	nama_bulan	nilai_seasonal
3	Apr	564,892,389
4	Mei	871,225,945
7	**Agt**	**909,720,939**
8	Sep	449,102,337
9	**Okt**	**1,364,499,178**
10	Nov	291,355,237

Low season:

	nama_bulan	nilai_seasonal
0	Jan	-5,690,792
1	Feb	-879,765,764
2	Mar	-443,475,246
5	**Jun**	**-968,215,288**
6	Jul	-322,689,228
11	**Des**	**-1,830,959,707**

Revenue Trend Analysis

The trend component indicates a structural growth phase in the early period, reaching a peak around the midpoint. Following this peak, the trend gradually declines, signaling a slowdown in core revenue momentum

Revenue Peak and Year-End Contraction

The seasonal analysis reveals a consistent annual pattern, with the **strongest uplift in October** (+1.36B above its normal level) and the **deepest decline in December** (–1.83B below its normal level).

This highlights significant year-end volatility and underscores the need for strategic demand and promotional planning during low-season periods.

Key Insights & Strategic Actions

Revenue is primarily volume-driven. MoM growth total of 1.1% signals weak momentum, influenced by recurring seasonal slowdowns. Channel, customer segment and provinces differences show no significant impact.

Optimize Seasonal Strategy

Strengthen performance during low-season months (June and December) in all segments/provinces, through targeted mid-year and year-end promotional programs

Maintain Unified Marketing Execution

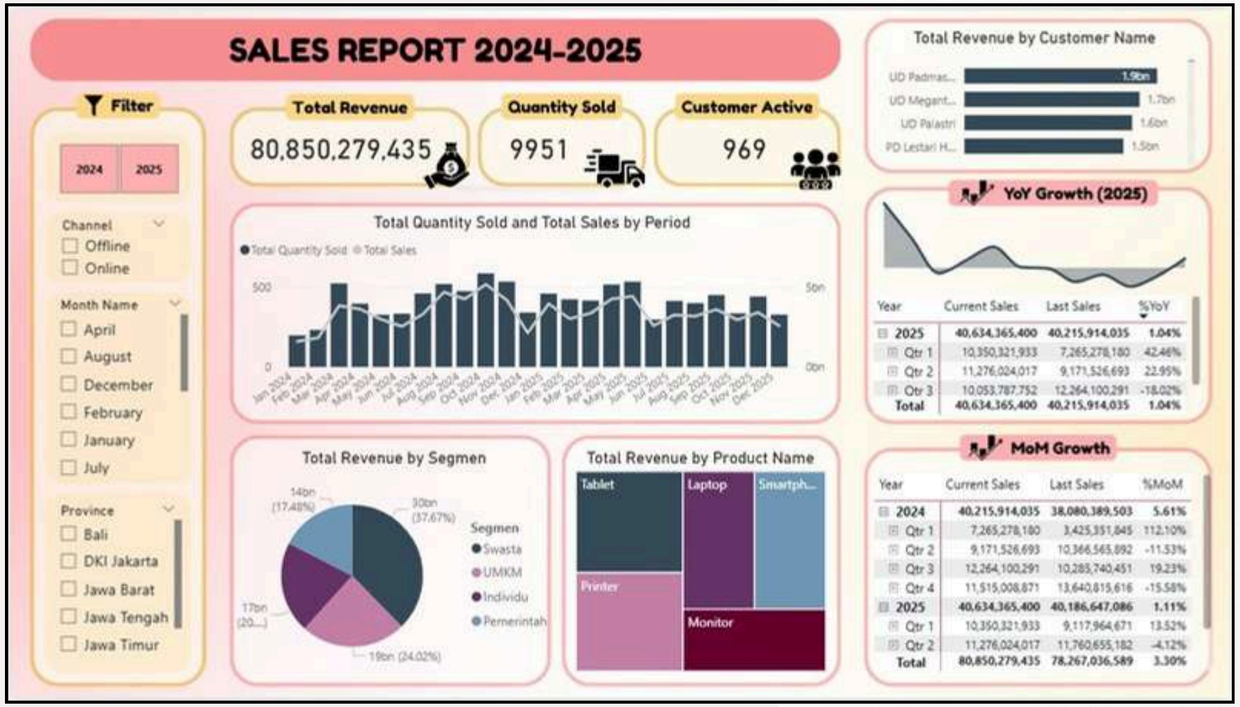
Continue standardized pricing and campaigns in all segments/provinces, across regions, strengthen offline execution suited for electronics, and preserve a balanced channel mix

Drive Volume Growth & Core Markets

Maintain stable pricing while driving volume through bundling and cross-selling, positioning DKI Jakarta as the growth engine and North Sumatra as the core revenue pillar

THANK YOU

Electronic Dashboard Preview



Project repository:
https://github.com/arindawardani02/electronic-sales-dashboard.git

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